



SUMMARY OF UPDATED OPTIMIZATION REPORT

March 25, 2026



AGENDA

Overall system performance

Current plume extents

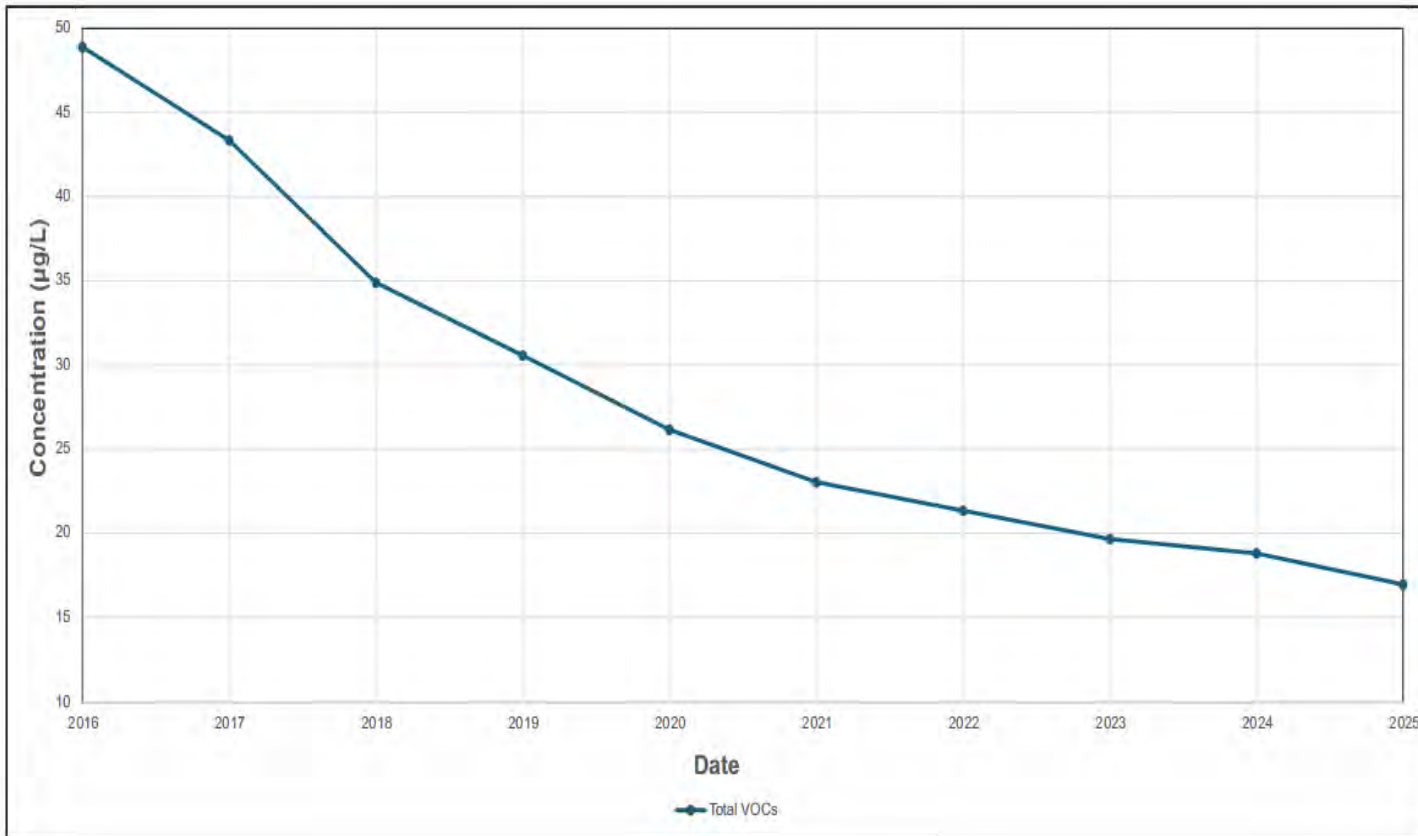
Extraction well TCE concentration trends

Rationale for extraction well shutdown

Potential next steps

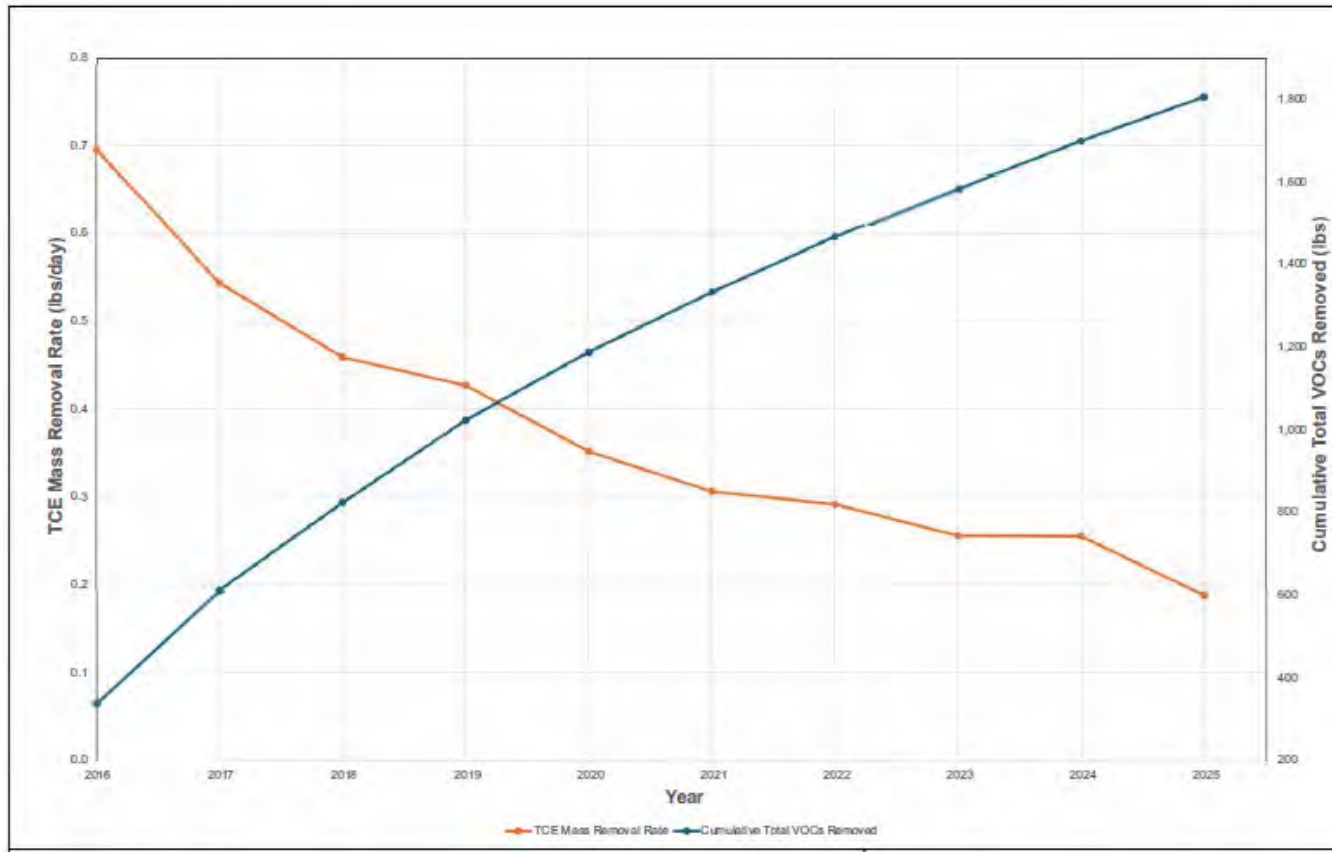
Other considerations

AVERAGE INFLUENT CONCENTRATION – TOTAL VOCs



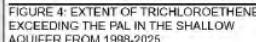
- Influent concentrations of total VOCs decreasing over time
- System continues to remove TCE/PCE mass
- May encounter diminishing returns scenario in near future

VOC MASS REMOVAL



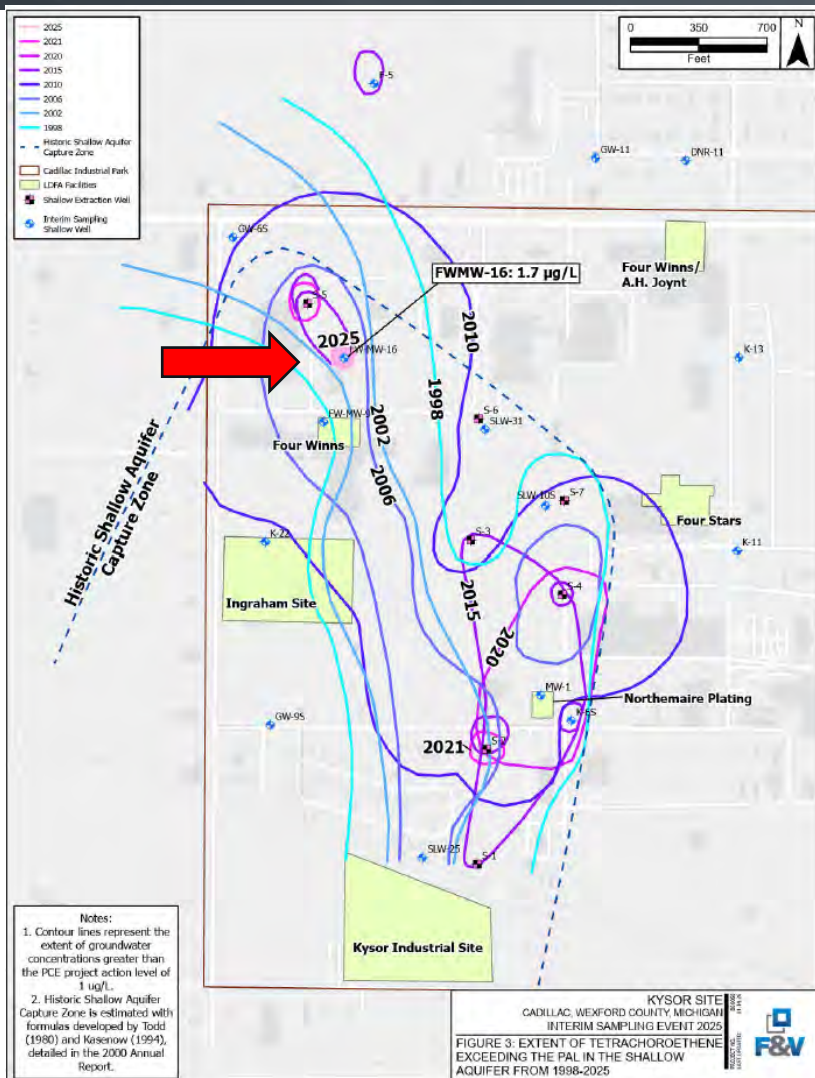
- Cumulative total beginning to flatten over time
- TCE/PCE mass removal rate decreasing over time
- May encounter diminishing returns scenario in near future

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TCE concentration in S-2 (5.4 µg/L) slightly exceeds the 5.0 µg/L TCL

SHALLOW PLUME EXTENT – PCE

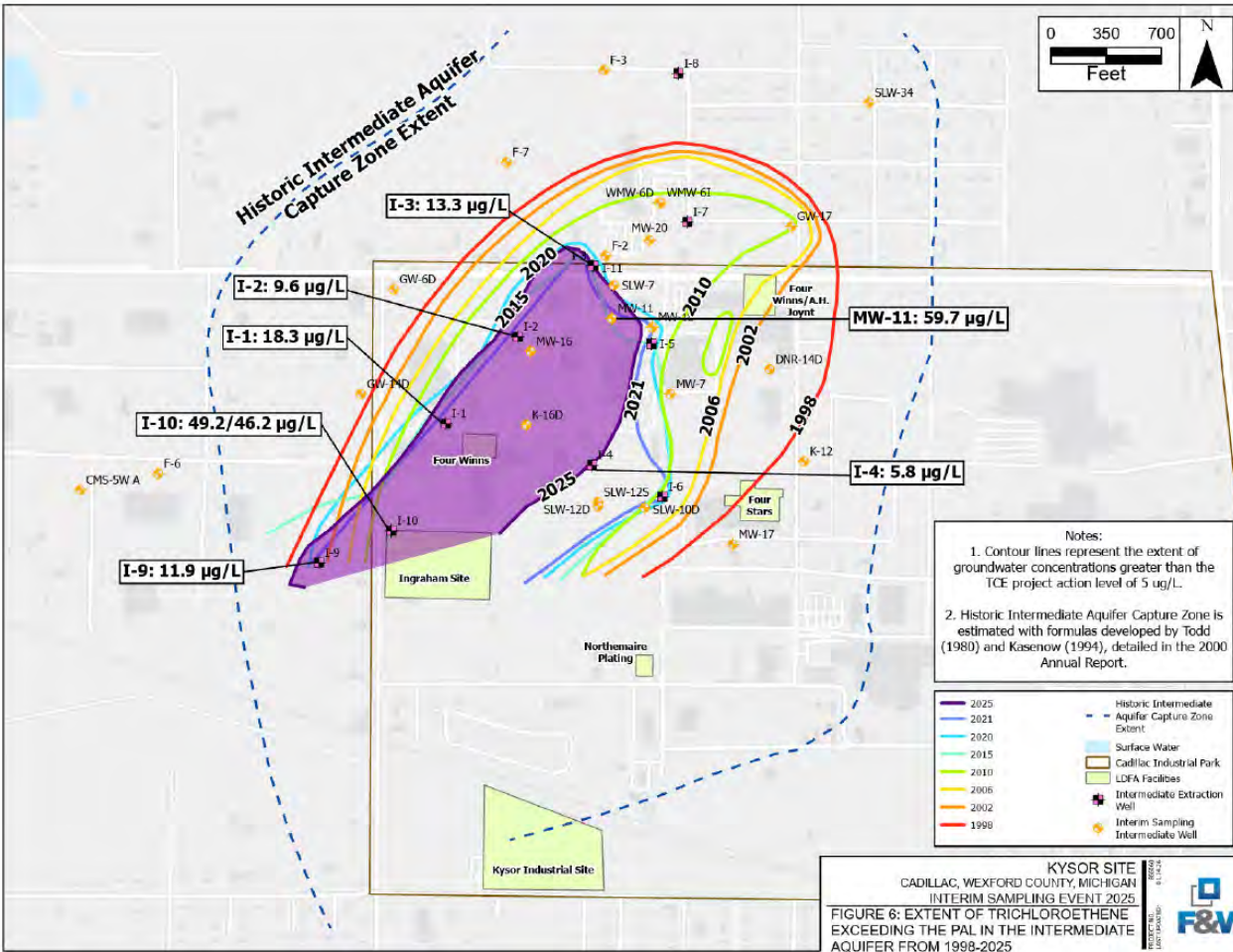


Plume footprint has been reduced to the vicinity of FWMW-16 (upgradient of S-5)

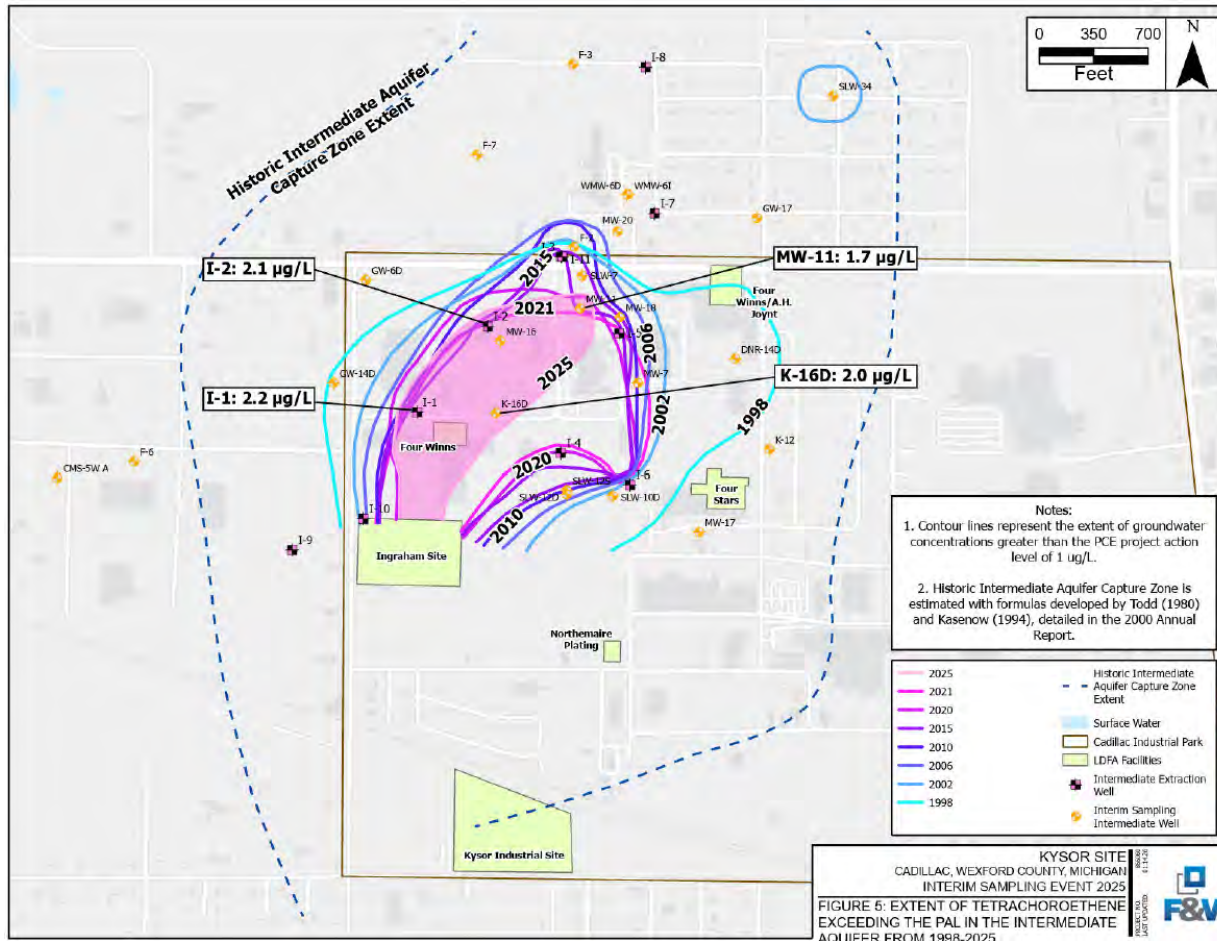
TCE concentration in FWMW-16 (1.7 µg/L) slightly exceeds the 1.0 µg/L TCL

Note: Data collected, analyzed, and presented by Fleis & VandenBrink

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INTERMEDIATE PLUME EXTENT – PCE



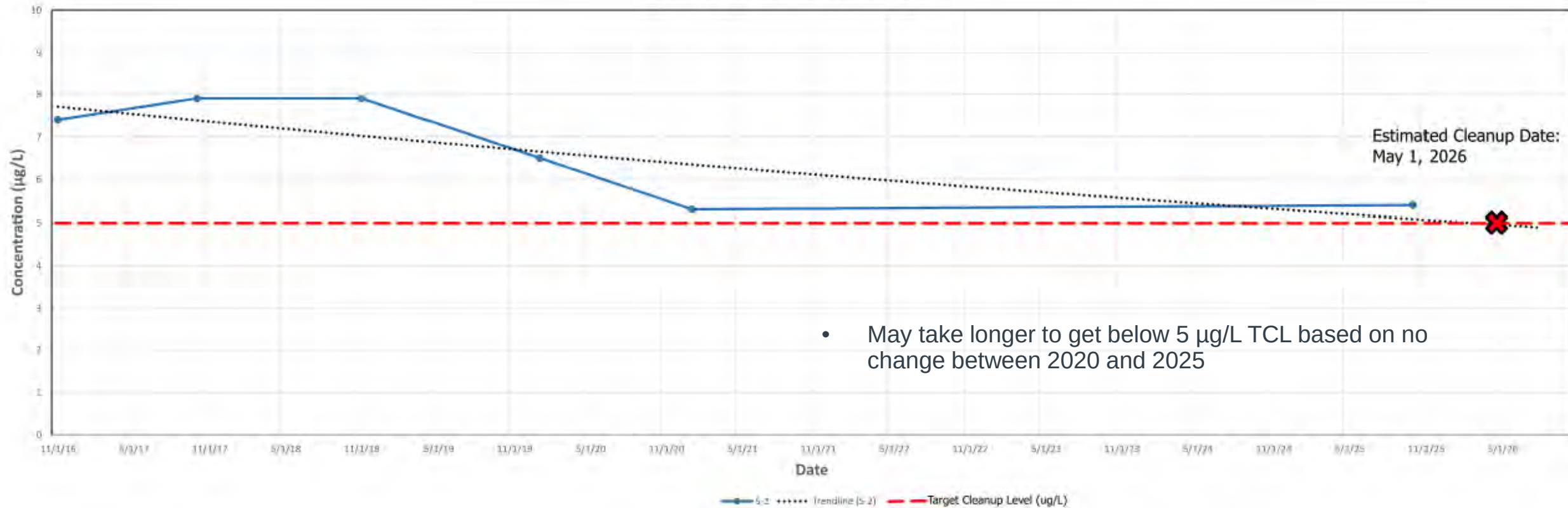
TCE TRENDS IN S-1, S-5, AND S-6

Graph 1: TCE Concentrations in S-1, S-5, & S-6



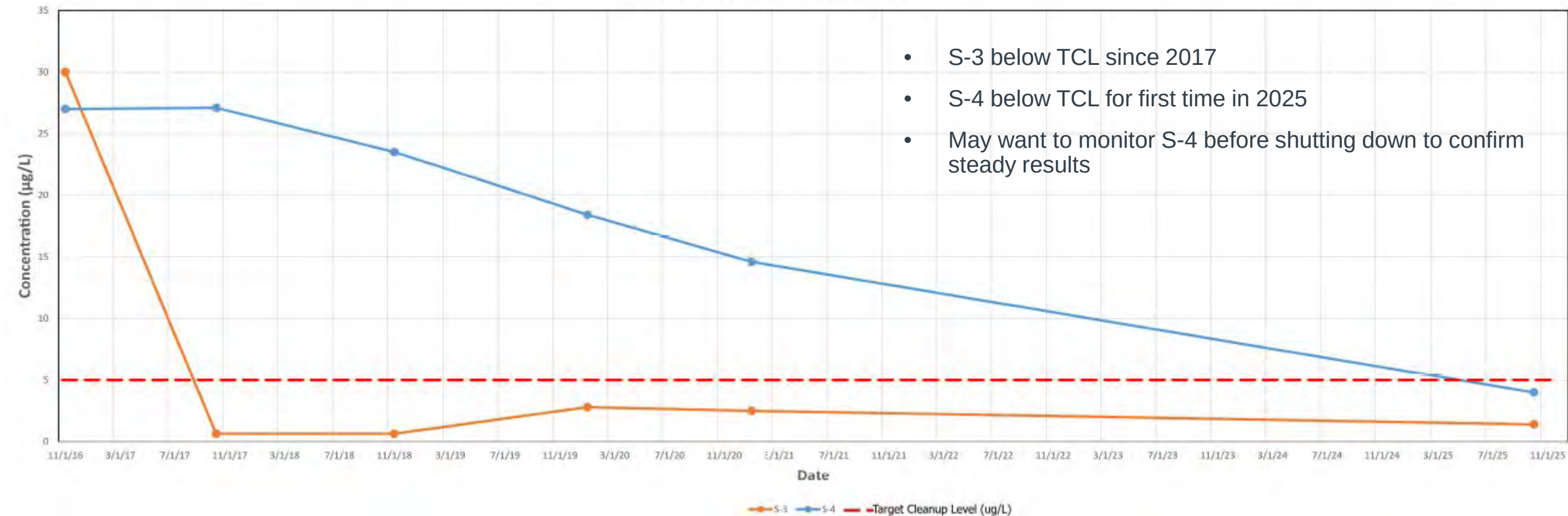
TCE TREND IN S-2

Graph 2: TCE Concentrations in S-2

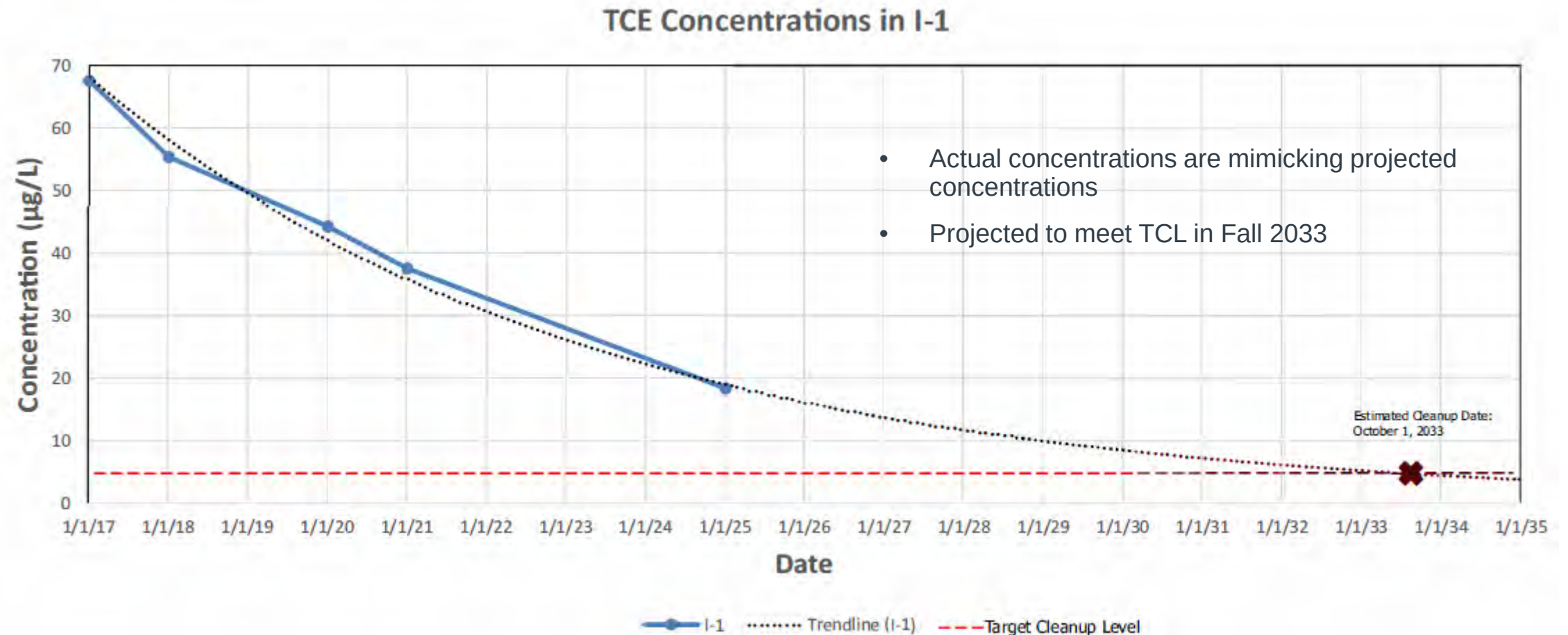


TCE TRENDS IN S-3 AND S-4

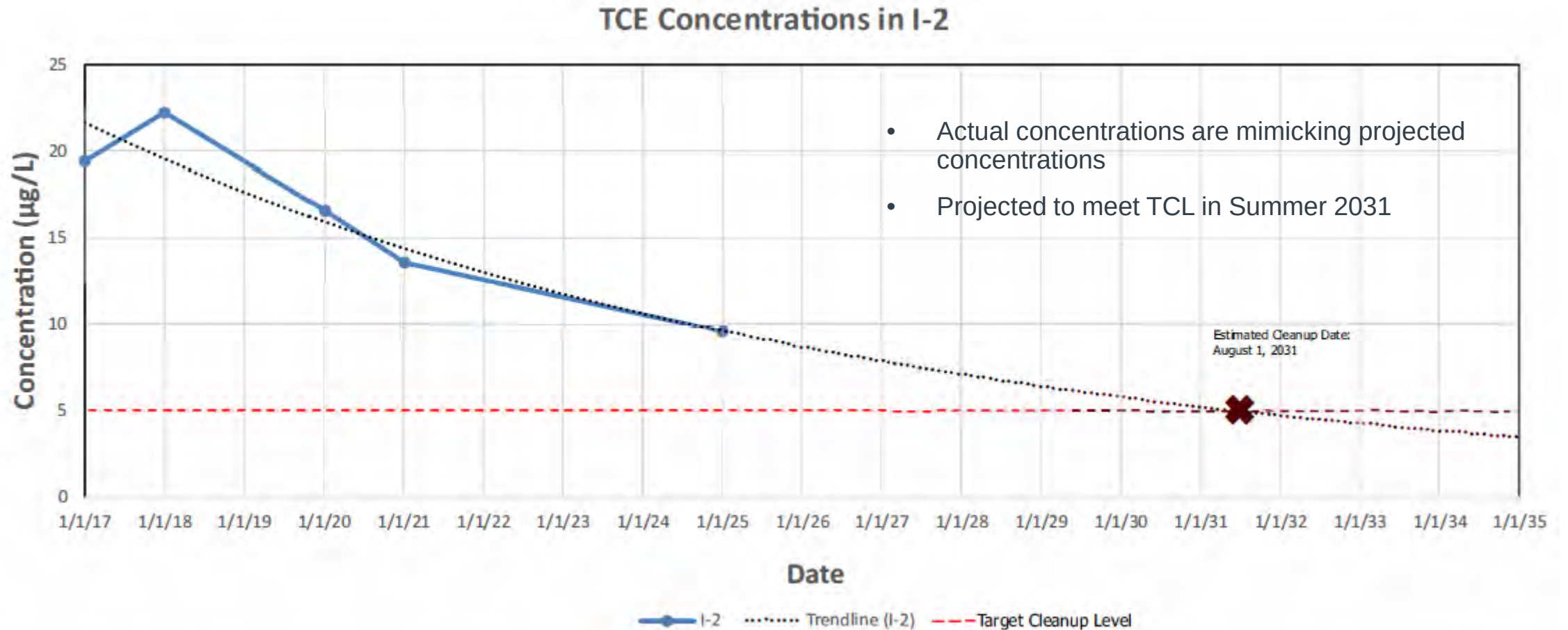
Graph 3: TCE Concentrations in S-3 & S-4



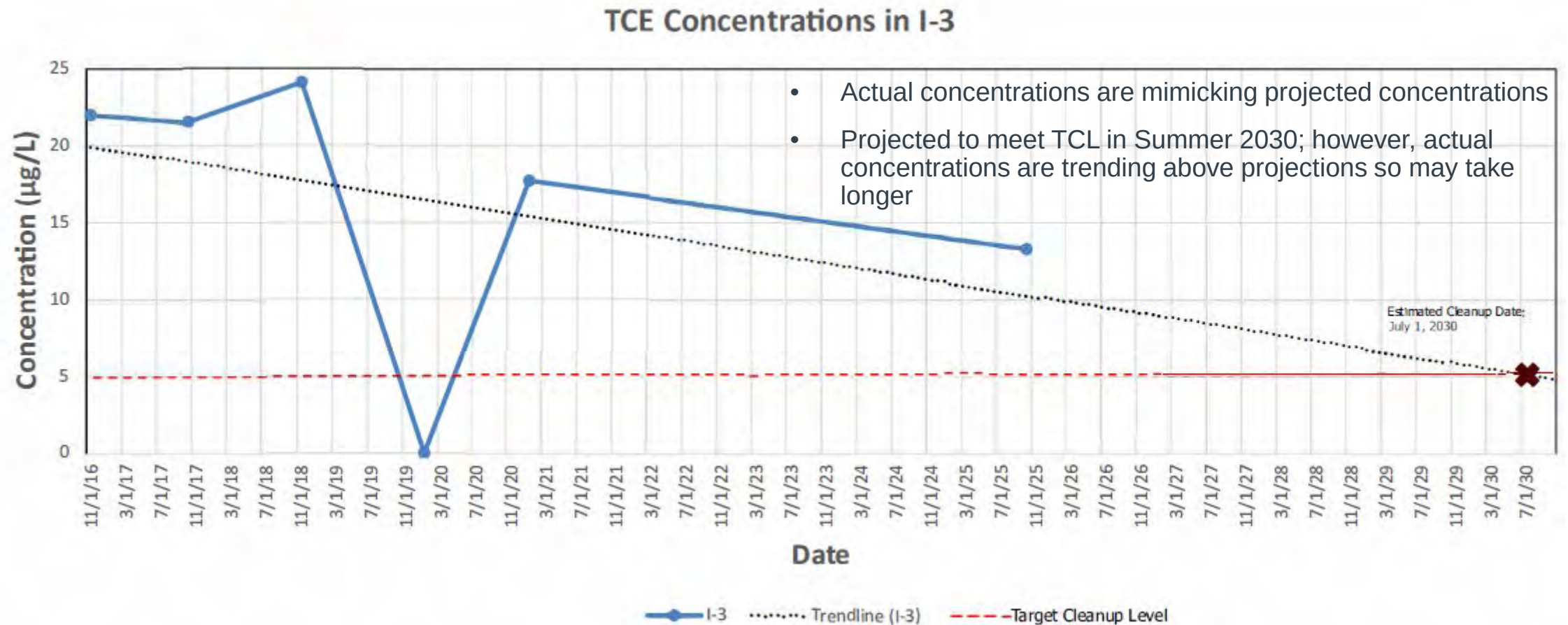
TCE TREND IN I-1



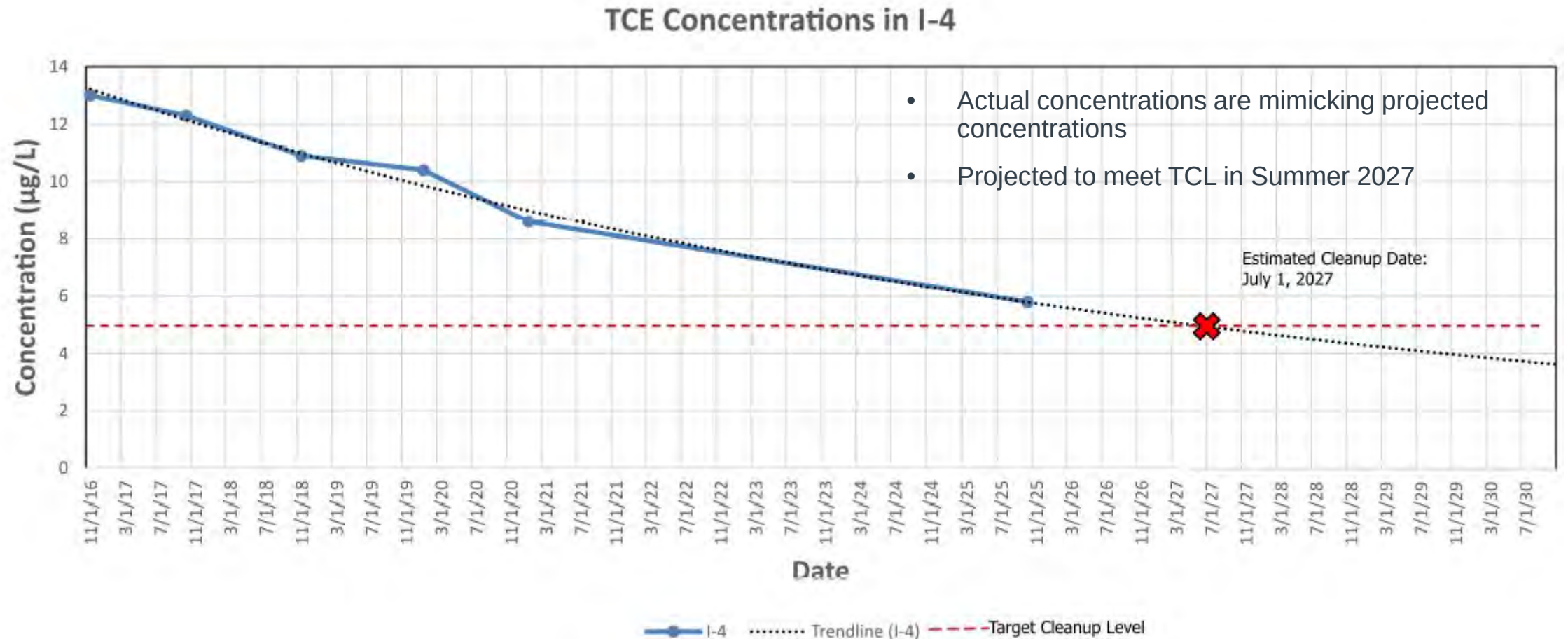
TCE TREND IN I-2



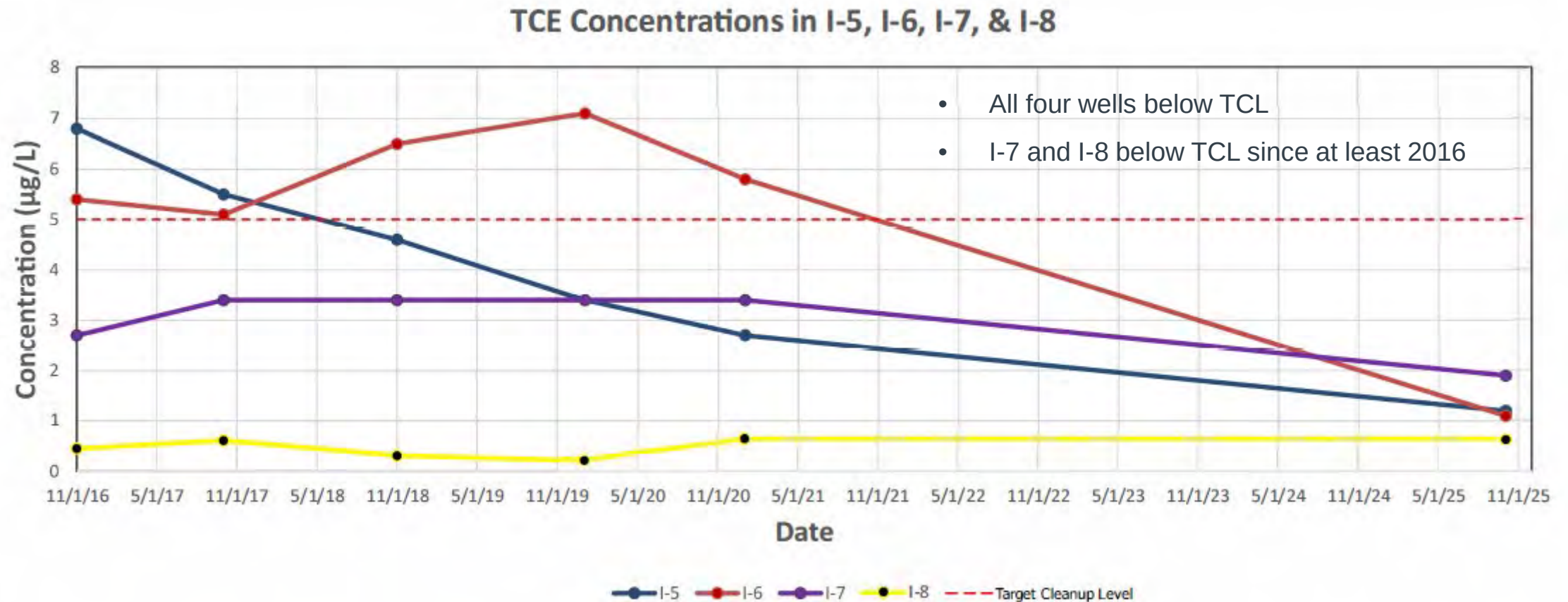
TCE TREND IN I-3



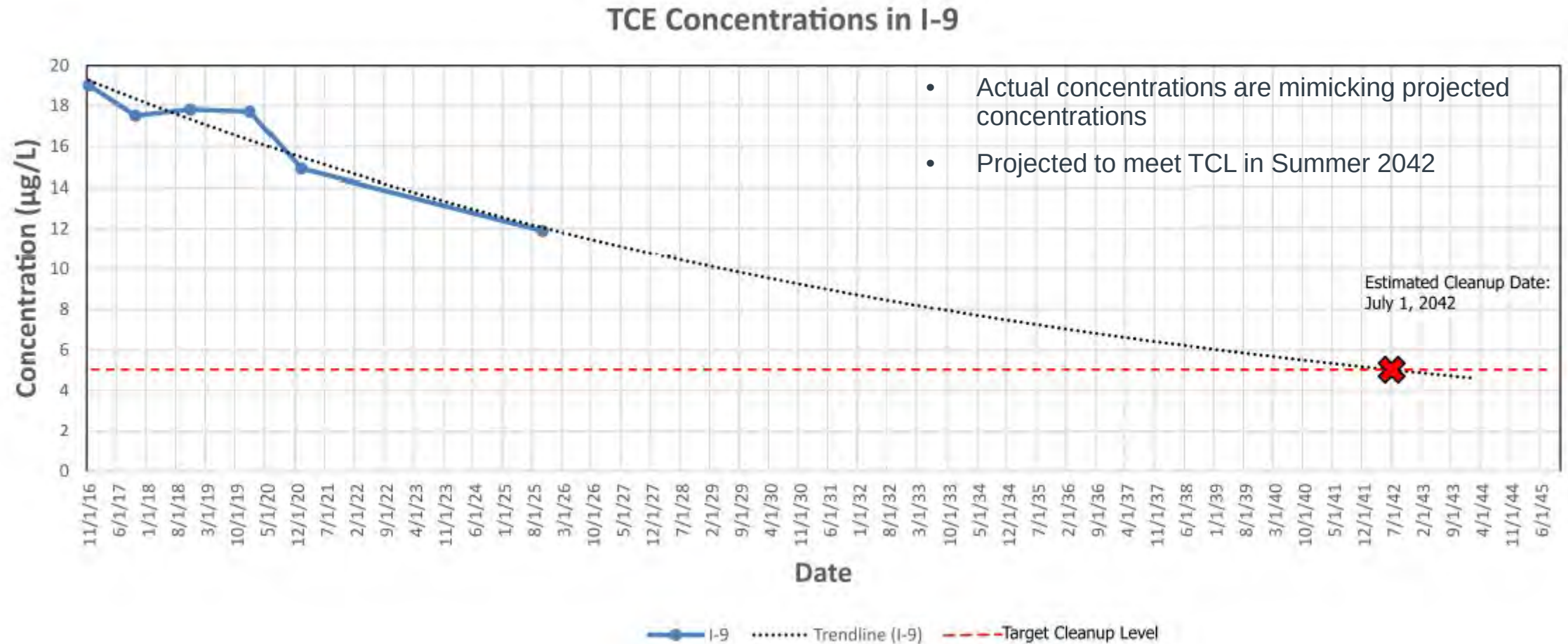
TCE TREND IN I-4



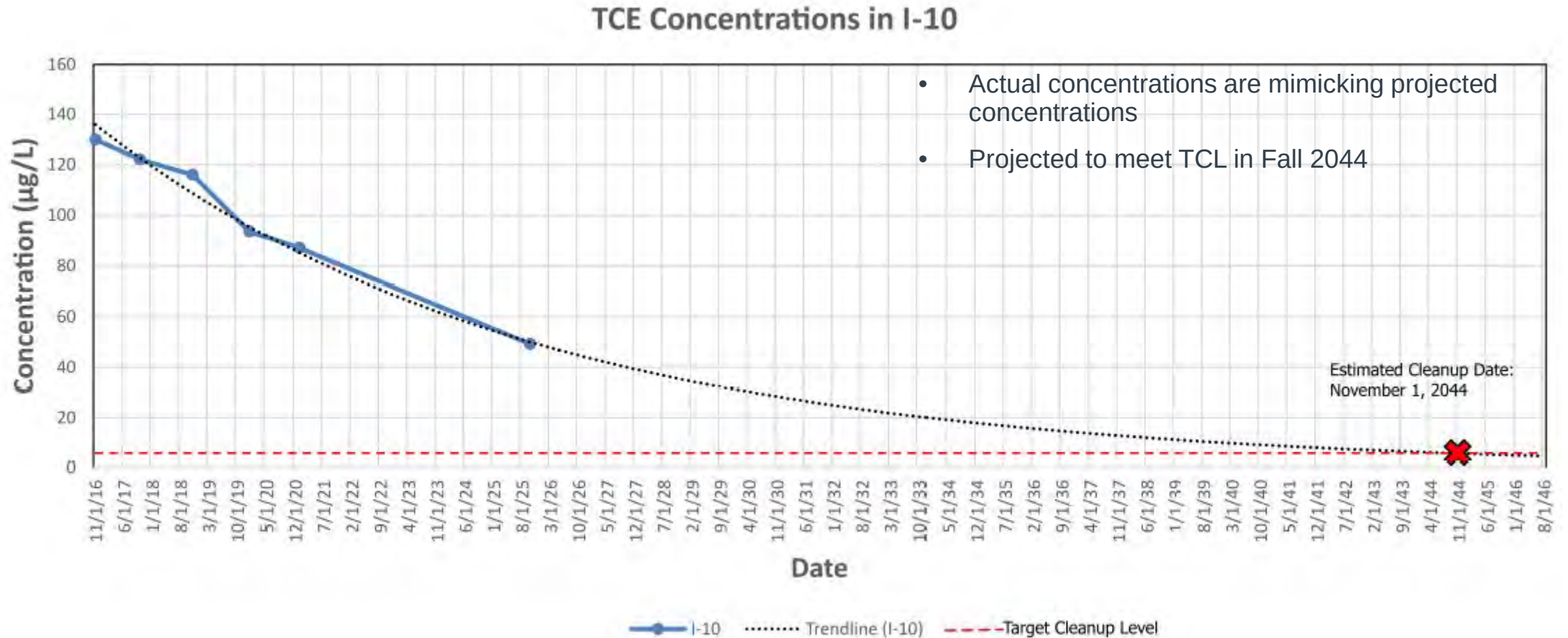
TCE TRENDS IN I-5, I-6, I-7, AND I-8



TCE TREND IN I-9



TCE TREND IN I-10



SUMMARY AND RECOMMEND ACTION

Extraction Well ID	Estimated Cleanup Date (TCE)
S-1	Currently below TCL
S-2	May 2026
S-3	Currently below TCL
S-4	Currently below TCL
S-5	Currently below TCL
S-6	Currently below TCL
I-1	October 2033
I-2	August 2031
I-3	July 2030
I-4	July 2027
I-5	Currently below TCL
I-6	Currently below TCL
I-7	Currently below TCL
I-8	Currently below TCL
I-9	July 2042
I-10	November 2044
I-11	Currently below TCL

- Only considers TCE concentration trends (primary driver)
- Presence of PFAS and 1,4-dioxane complicates the issue and may extend remedial objectives
- Recommend shutdown of S-3, I-7, and I-8
- Recommend further evaluating S-4 and I-6 for possible shutdown – likely will need more data to demonstrate repeatable results

EXTRACTION WELL SHUTDOWN RATIONALE

SHALLOW AQUIFER

Well ID	PCE and/or TCE	PFAS (2024)	1,4-Dioxane
S-1			
S-2			
S-3			
S-4			
S-5			
S-6			
S-7	-	-	-

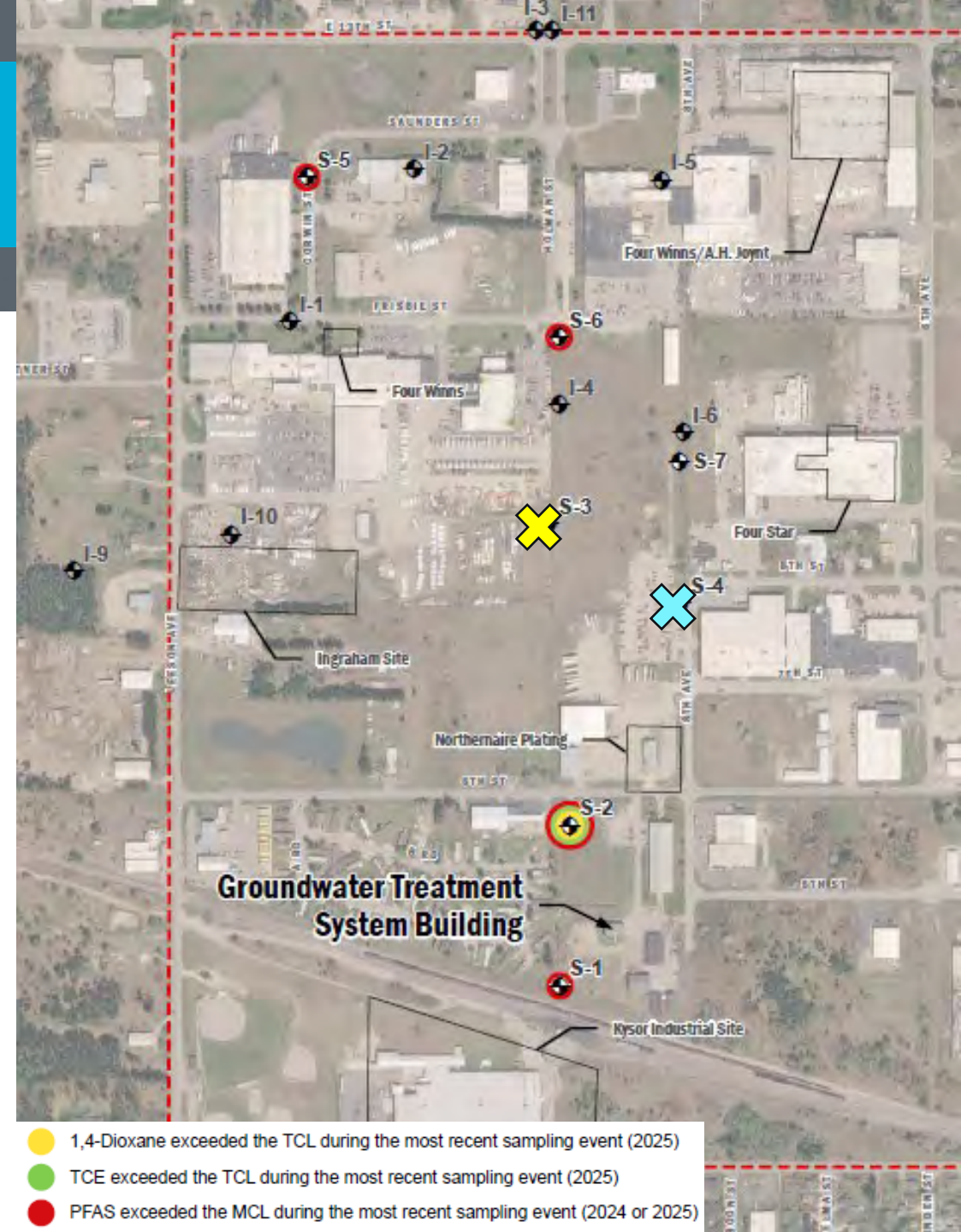
● = Below TCL or other criterion

● = Close to exceeding TCL or other criterion

● = Above TCL or other criterion

 = Recommended for shutdown

 = Monitor for possible shutdown



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● 1,4-Dioxane exceeded the TCL during the most recent sampling event (2025)

- TCE exceeded the TCL during the most recent sampling event (2025)

● PFAS exceeded the MCL during the most recent sampling event (2024 or 2025)

INTERMEDIATE AQUIFER

Well ID	PCE and/or TCE	PFAS (I-1 – I-4 2025) (I-5 – I-11 2024)	1,4-Dioxane
I-1	●	●	●
I-2	●	●	●
I-3	●	●	●
I-4	●	● ●	●
I-5	●	●	●
I-6	●	● ●	●
I-7	●	● ●	●
I-8	●	● ●	●
I-9	●	● ●	●
I-10	●	●	●
I-11	●	●	●

● = Below TCL or other criterion

● = Close to exceeding TCL or other criterion

● = Above TCL or other criterion

✕ = Recommended for shutdown

✕ = Monitor for possible shutdown



● PCE and/or TCE exceeded the TCL during the most recent sampling event (2025)

● PFAS exceeded the MCL during the most recent sampling event (2024 or 2025)

PFAS - SHALLOW AQUIFER

Well ID	EPA MCLs	EGLE Part 201	EGLE Surface Water Values
S-1	●	● ●	●
S-2	●	●	●
S-3	● ●	●	●
S-4	● ●	●	●
S-5	●	●	●
S-6	●	●	●
S-7	-	-	-

- = Below TCL or other criterion
- = Close to exceeding TCL or other criterion
- = Above TCL or other criterion

Note:
 PFAS data from 2025 for I-1 through I-4
 PFAS data from 2024 for I-5 through I-11



PFAS – INTERMEDIATE AQUIFER

Well ID	EPA MCLs	EGLE Part 201	EGLE Surface Water Values
I-1	●	●	●
I-2	●	●	●
I-3	●	●	●
I-4	● ●	●	●
I-5	●	●	●
I-6	● ●	●	●
I-7	● ●	●	●
I-8	● ●	●	●
I-9	● ●	●	●
I-10	●	●	●
I-11	●	●	●

● = Below TCL or other criterion

● = Close to exceeding TCL or other criterion

● = Above TCL or other criterion

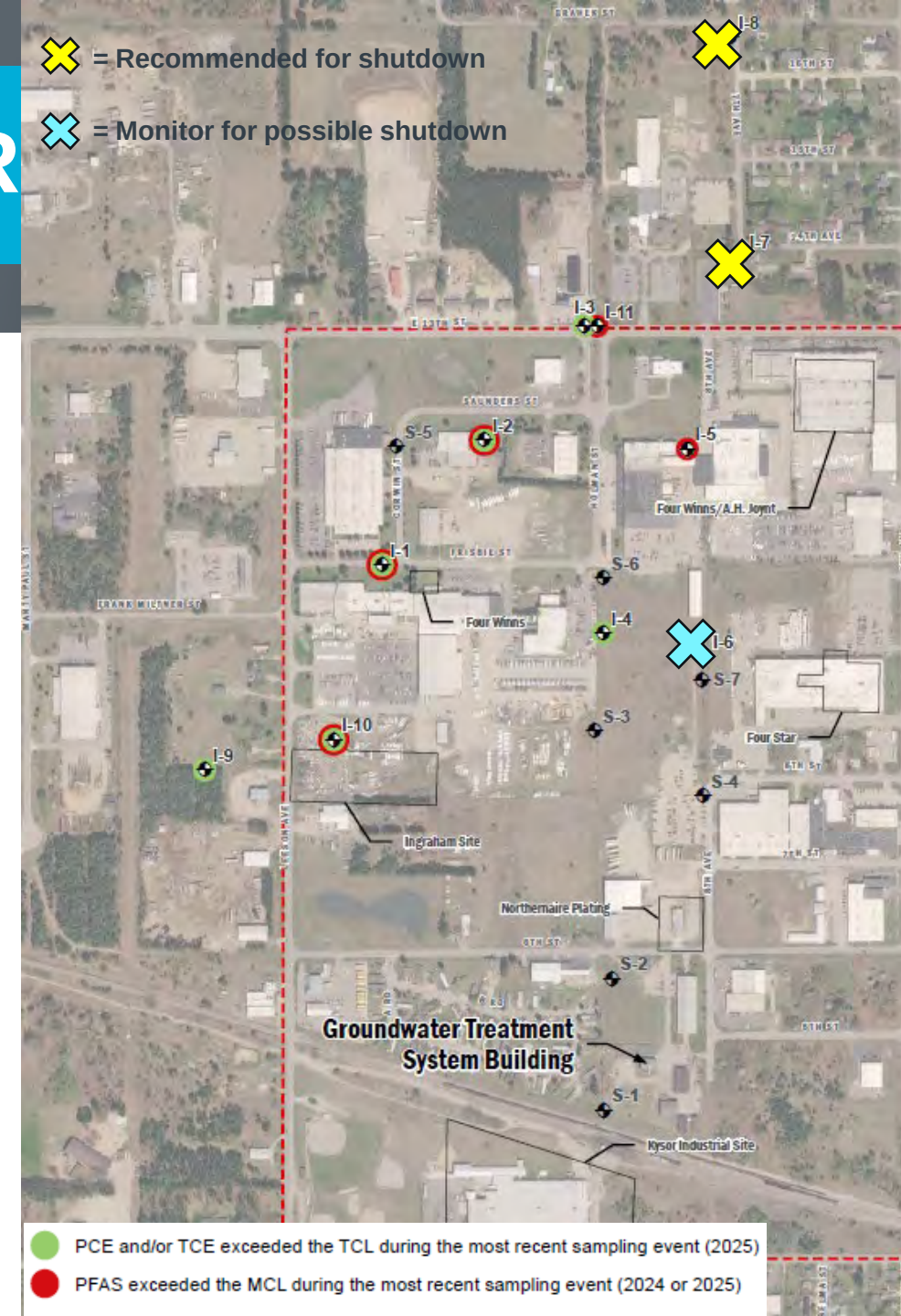
Note:

PFAS data from 2025 for I-1 through I-4

PFAS data from 2024 for I-5 through I-11

✖ = Recommended for shutdown

✖ = Monitor for possible shutdown



- PCE and/or TCE exceeded the TCL during the most recent sampling event (2025)
- PFAS exceeded the MCL during the most recent sampling event (2024 or 2025)

POTENTIAL NEXT STEPS

For LDFA Consideration:

Develop a Shutdown Strategy to Present to EPA

- Shutdown extraction wells S-3, I-7, and I-8
- Monitor concentrations in vicinity of S-3, I-7, and I-8
- Develop decision criteria/risk tolerance for shutdown of S-4 and I-6 and others

Evaluate Impact of PFAS and 1,4-Dioxane on Remedial Objectives

- Uncertainty exists; confirm LDFA's regulatory obligations.
- Are PFAS and 1,4-dioxane considered COCs for the site?

Carbon Vessel Breakthrough Modeling

- Would aid in short- and long-term budgetary and operational planning, especially if additional wells need to be added to influent.
- Need to better understand PFAS treatment needs.

Development of an O&M Plan for the Carbon Vessel(s)

- Would be helpful following breakthrough modeling.
- Adding additional extraction wells to influent will impact use and longevity of the GAC.

Consider an Energy Efficiency Audit of Existing Equipment

- Are there further savings if the air stripper blower was updated, for example?

BIG PICTURE CONSIDERATIONS

OTHER CONSIDERATIONS

Regulatory Uncertainty

- EPA indicates original exposure assumptions, toxicity data, cleanup levels, and remedial action objectives are not valid (page 27).
- Focused on PFAS and 1,4-dioxane
- Open-ended

Remedial Objectives and Obligations

- What has changed since the ROD?
- Is the LDFA obligated to capture and treat PFAS and 1,4-dioxane?
- Is EPA required to formally add one or both as a COC via ROD amendment or ESD?

Groundwater Modeling

- Capture analysis needed to evaluate downgradient exposure risks?
- Would it be beneficial to add extraction wells to reduce remedial time frame?
- Update original modeling?

WHAT ARE THE LDFA'S PFAS REGULATORY OBLIGATIONS?

From the Sixth Five-Year Review (August 13, 2025):

Page 21-22

“The LDFA also sampled all the LDFA treatment plant extraction wells and the treatment plant influent and effluent for PFAS.

Based on the results of this sampling, it appeared the Kysor and Northernaire Sites were not the likely source of PFAS. At this time, EGLE and MPART consider the MPART investigation area associated with the industrial park complete.”

Page 28

“Results from the MPART investigation identified PFAS both side-gradient and upgradient from the Site in exceedance of EGLE’s and/or the EPA’s PFAS MCLs. The presence of PFAS at these locations indicates the Site is not the primary source of PFAS contamination in the surrounding residential wells.”

Page 28-29

“The EPA anticipates that the fall 2025 sampling event will confirm conclusions from the August 2024 Cadillac Area of Interest PFAS investigation by MPART that the Site is not the primary source of PFAS contamination in surrounding residential wells.”

SITE-WIDE QUESTIONS



2025 Five-Year Review Raises Concerns

Have these been identified and discussed?

Are the remedial objectives changing?

What control do you have?



What is Your Regulatory Strategy?

Groundwater monitoring and treatment system strategies should align with broader objectives.

Does the LDFA have an experienced advocate to lead?

A proactive approach is often more effective than a reactive approach.



What Potential Risks Exist?

Is there an upcoming ROD amendment or ESD?

Are treatment system changes needed?

What are your off ramps?



Revised Groundwater Monitoring Plan

What is it based on and is it justified?

How does it impact short- and long-term remedial objectives?

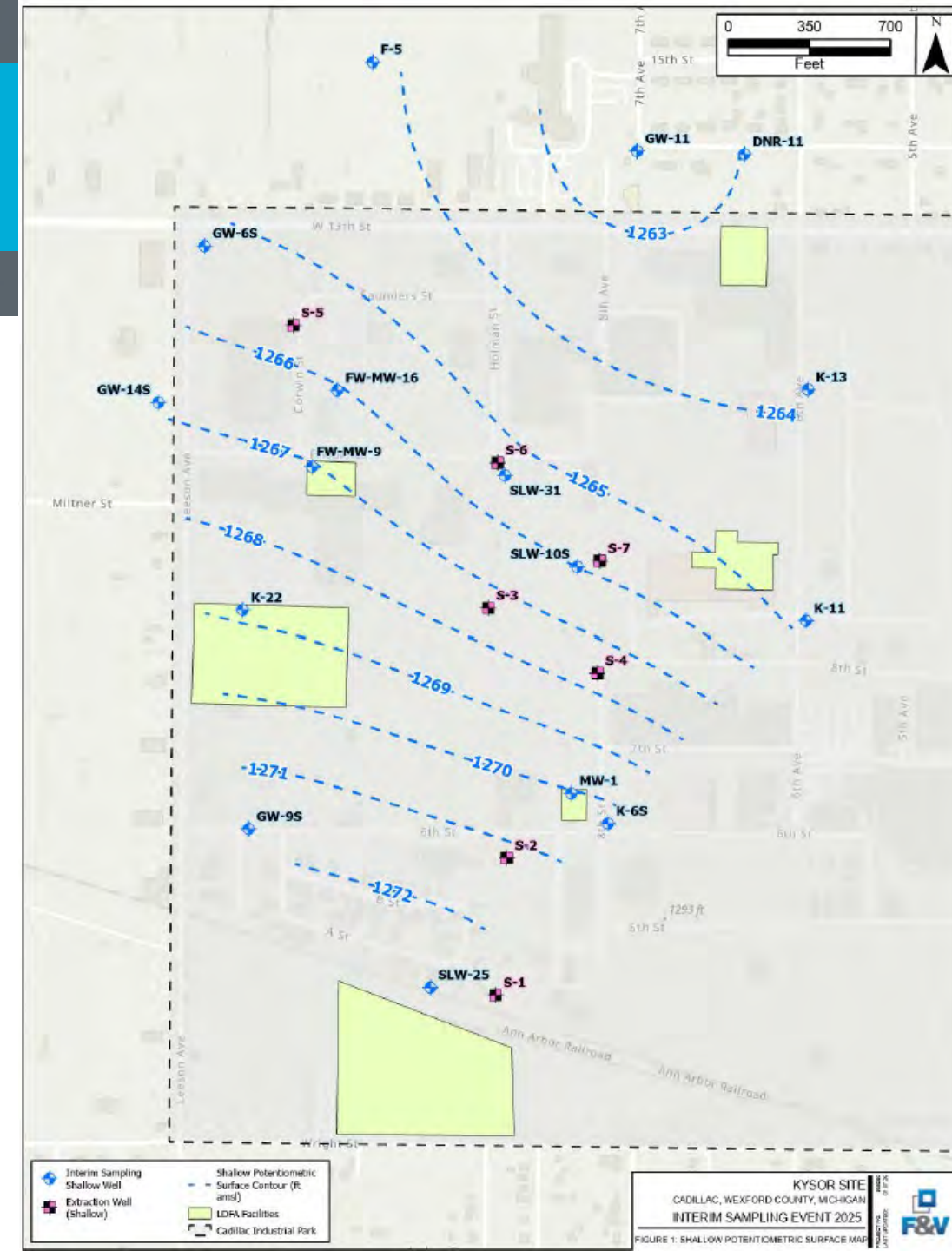
Would a revised conceptual site model aid in challenging the changes?

DISCUSSION

Matt Vander Eide, PG, CPG
616-481-5771
matthew.vandereide@geosyntec.com

GROUNDWATER FLOW

- Shallow aquifer does not appear to be heavily influenced by extraction wells
- Flows north/northeast at 200 ft/yr (per the ROD)



- Intermediate aquifer appears to be influenced by extraction wells
- Flows north/northwest at 80 ft/yr (per the ROD)

